

INFO 7225

Financial Accounting

ACCOUNTING FOR LONG-TERM ASSETS

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The Basics of Financial Accounting:

Special Topic 3:

Accounting for Long-term/Fixed Assets

Fixed-Asset Management

Even small organizations generally own many fixed assets, which management must track as they are purchased and used. The objective of the fixed-asset management (FAM) function is to manage the purchase, maintenance, valuation, and disposal of an organization's fixed assets (also called "long-term assets" because they last more than 1 year).

In thinking about how complex it might be to track fixed assets, imagine the endless number of fixed assets that are owned by a large, publicly traded firm such as Boeing (which has about \$12,807 million in net property, plant, and equipment on its 2016 balance sheet). A firm must record each fixed asset on its books when it purchases the asset. In addition, the firm must maintain depreciation schedules for its fixed assets. Many large firms will calculate depreciation using five or more different depreciation methods because of different calculation requirements for GAAP financial statements, federal

income tax reporting, state tax reporting, and so forth. Employees often move fixed assets around within an organization, and an effective AIS should keep track of all asset locations. Bar codes affixed to physical assets make this job easier.

Because fixed assets often require repairs, an AIS should also track repair costs and distinguish between revenue expenditures and capital expenditures. Revenue expenditures are ordinary repair expenses, whereas capital expenditures add to the value of assets. Finally, the AIS calculates the amount of gain or loss upon disposal of individual fixed assets. By comparing the amount received for the asset with the asset's book value, the AIS can compute a gain or loss. Fortunately, software companies offer a variety of solutions to help managers and automate these processes.

LEARNING OBJECTIVES

After completing this session, you should be able to

- 1. Distinguish between tangible and intangible assets;**
- 2. Analyze and classify capitalized costs versus expenses;**
- 3. Explain and apply different depreciation methods to allocate capitalized costs;**
- 4. Describe accounting for intangible assets and record related transactions;**
- 5. Describe some special issues in accounting for long-term assets.**

Distinguish between Tangible and Intangible Assets

- Assets are items a business owns.
- For accounting purposes, assets are categorized as *current* versus *long term*, and *tangible* versus *intangible*.
- Long-term assets
 - Assets that are expected to be used by the business for more than one year are considered long-term assets.
 - They are **not** intended for resale and are anticipated to help generate revenue for the business in the future.
 - Some common long-term assets are computers and other office machines, buildings, vehicles, software, computer code, and copyrights.
 - Although these are all considered long-term assets, some are tangible and some are intangible.

Tangible Assets

- An asset is considered a tangible asset when it is an economic resource that has physical substance—it can be seen and touched.

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graph TD; A[An asset is considered a tangible asset when it is an economic resource that has physical substance—it can be seen and touched.] --- B[Short term, such as inventory and supplies]; A --- C[Long term, such as land, buildings, and equipment (aka fixed asset)]; C --> D[1) Be used in the normal operation of the business for more than one year<br/>2) not near the end of its useful life, and<br/>3) the company must have no plan to sell the item in the near future];
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Short term, such as inventory and supplies

Long term, such as land, buildings, and equipment (aka fixed asset)

- 1) Be used in the normal operation of the business for more than one year
- 2) not near the end of its useful life, and
- 3) the company must have no plan to sell the item in the near future

Intangible Assets

- Companies may have other long-term assets used in the operations of the business that they do not intend to sell, but that do not have physical substance; these assets still provide specific rights to the owner and are called intangible assets.
- These assets typically appear on the balance sheet following long-term tangible assets.
- Examples of intangible assets:
 - Patents, copyrights, franchises, licenses, goodwill, sometimes software, and trademarks.
- Because the value of intangible assets is very subjective, it is usually **not** shown on the balance sheet until there is an event that indicates value objectively, such as the purchase of an intangible asset.

Useful Life of Intangible Assets

Asset	Useful Life
Patents	Twenty years
Trademarks	Renewable every ten years
Copyrights	Seventy years beyond death of creator
Goodwill	Indefinite

APPLE INC.
Consolidated Balance Sheets
(in millions)

Assets	2017	2016
Current Assets:		
Cash and Cash Equivalents	\$ 20,289	\$ 20,484
Short-term Marketable Securities	53,892	46,671
Accounts Receivable, Allowances of \$58 and \$53, respectively	17,874	15,754
Inventories	4,855	2,132
Vendor Nontrade Receivables	17,799	13,545
Other Current Assets	13,936	8,283
Total Current Assets	128,645	106,869
Long-term Marketable Securities	194,714	170,430
Property, Plant, and Equipment, net	33,783	27,010
Goodwill	5,717	5,414
Acquired Intangible Assets, net	2,298	3,206
Other Noncurrent Assets	10,162	8,757
Total Assets	<u>\$375,319</u>	<u>\$321,686</u>

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4. Describe accounting for intangible assets and record related transactions;
5. Describe some special issues in accounting for long-term assets.

Important Concepts: Expense vs. Depreciation

Two types of business expenditures

- **EXPENSES:** Expenditures such as labor, services, materials are examples of items that are fully deducted from current year taxable income;
- **DEPRECIATION:** Expenditures for capital assets, on the other hand, cannot be fully deducted from taxable income in the year in which they occur; instead, they must be spread out or distributed over some allowable recovery period.

Capitalized Costs on PPE

Property, plant, and equipment (PPE)

- Used in the business operations
- Useful life > 1 year

The asset is typically capitalized.  (“Capitalization”)

Matching principle

The historical cost of the long-term asset is recorded on the balance sheet when the asset is acquired.

Its allocated costs are expensed on the income statement over *the asset’s economic life*.

Distinguish between PPE and Investment

When a business purchases a long-term asset (used for more than one year), it classifies the asset based on whether the asset is used in the business's operations.

Property, plant, and equipment (PPE)

- **Used in the business operations.**
- For example, if a business owns land on which it operates a store, warehouse, factory, or offices, the cost of that land would be included in property, plant, and equipment.

Investment

- Long-term assets that are **not** used in daily operations
- If a business owns a vacant piece of land on which the business conducts no operations (and assuming no current or intermediate-term plans for development), the land would be considered an investment.

Distinguish between PPE and Investment

A long-term tangible asset is acquired.



Used in the day-to-day operations of the business?

Yes

The cost is capitalized and then depreciated over the useful life of that asset.

No

Purchased for investment purposes; will be considered an investment asset:

- short term (can be converted to cash in one year) or
- long term (held for over a year)

Costs Incurred on Service of PPE

Is the useful life of the
asset extended?



Yes

Capitalized and then depreciated.

No

Expensed during current reporting period.

Expenditures Are Either Capitalized Or Expensed

1. Assets: PPE or investment?

- 1) Land next to the production facility held for use next year as a place to build a warehouse _____
- 2) Land held for future resale when the value increases _____
- 3) Equipment used in the production process _____

2. Expenditures: capitalized or expensed?

- 1) Normal repair and maintenance on the manufacturing facility _____
- 2) Cost of taxes on new equipment used in business operations _____
- 3) Shipping costs on new equipment used in business operations _____
- 4) Cost of a minor repair on existing equipment used in business operations _____

Analyze and Classify Capitalized Costs versus Expenses

- Capitalization is the process by which
 - a long-term asset is recorded on the balance sheet, and
 - its allocated costs are expensed on the income statement over *the asset's economic life*.

JOURNAL			
Date	Account	Debit	Credit
Jan. 1, 2019	Machine Cash	5,000	5,000

JOURNAL			
Date	Account	Debit	Credit

LEARNING OBJECTIVES

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5. Describe some special issues in accounting for long-term assets.

Apply Depreciation Methods to Allocate Capitalized Costs

A long-term asset is acquired.



The historical (initial) cost, including any costs to acquire the asset and get it ready for use, is recorded on B/S (“capitalization”)



The capitalized cost is allocated over its anticipated economic (useful) life → I/S

There are 3 methods to allocate the cost.

Depreciation: the process of allocating the cost of using a long-term asset over its economic life.

Three Methods to Calculate Depreciation

Depreciation Method	Calculation
Straight line	$(\text{Cost} - \text{salvage value}) / \text{Useful life}$
Units of production	$(\text{Cost} - \text{salvage value}) \times (\text{Units produced in current period} / \text{Estimated total units to be produced})$
Double declining balance	$\text{Book value} \times \text{Straight-line annual depreciation percentage} \times 2$

Terminologies

The following items are important in determining and recording depreciation:

- **Cost basis**
 - Historical (initial) cost of the fixed asset, including any costs to acquire the asset and get it ready for use
- **Salvage (residual) value**
 - The price the asset will sell for or be worth as a trade-in when its useful life expires.
 - Often, the salvage value is estimated based on past experiences with similar assets
- **Useful life**
 - The length of time the asset will be productively used within operations
- **Book value**
 - The asset's original cost less accumulated depreciation.

Three Methods to Calculate Depreciation

Cost basis = P; salvage value = F; useful life = n

Straight-line method

Units-of-production method

Double-declining-balance method

Calculating and Recording Depreciation Costs

- Liam buys his silk screen machine for \$10,000. He estimates that he can use this machine for five years or 100,000 presses, and that the machine will only be worth \$1,000 at the end of its life. He also estimates that he will make 20,000 clothing items in year one and 30,000 clothing items in year two. Determine Liam's depreciation costs for his first two years of business using the following method and record the journal entries:
 - straight-line method
 - units-of-production method, and
 - double-declining-balance method

Calculating and Recording Depreciation Costs

- Liam buys his silk screen machine for \$10,000. He estimates that he can use this machine for five years or 100,000 presses, and that the machine will only be worth \$1,000 at the end of its life. He also estimates that he will make 20,000 clothing items in year one and 30,000 clothing items in year two. Determine Liam's depreciation costs for his first two years of business using the following method and record the journal entries:

➤ **Straight-line method**

JOURNAL			
Date	Account	Debit	Credit

Calculating and Recording Depreciation Costs

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 - **Units-of-production method**

Calculating and Recording Depreciation Costs

- Liam buys his silk screen machine for \$10,000. He estimates that he can use this machine for five years or 100,000 presses, and that the machine will only be worth \$1,000 at the end of its life. He also estimates that he will make 20,000 clothing items in year one and 30,000 clothing items in year two. Determine Liam's depreciation costs for his first two years of business using the following method and record the journal entries:
 - **Double declining balance method**

Double-declining-balance (DDB) Method

- Declining percentage $p =$
- DDB, $p =$
- Two features:
 1. Salvage value is not used in calculation of depreciation expense
 2. Assets are not allowed to depreciate below their SVs

Year	Depreciation Expense	Accumulated Depreciation	Book Value

Allocate Costs for Natural Resources and Intangible Assets

PPE (long-term tangible; fixed assets)



Depreciation

Natural resources:

- Tangible assets occurring in nature that a company owns, which are consumed when used



Depletion

- Natural resources are depleted over the life of the asset
- Use a units-consumed method
- Depletion; contra account: accumulated depletion

Intangible assets:

Amortized over the life of the asset



Amortization

- No salvage value
- Straight-line method
- No contra account required

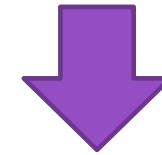
Fundamentals of Depletion of Natural Resources

Natural resources:

- Tangible assets occurring in nature that a company owns, which are consumed when used
- Examples include lumber, mineral deposits, and oil/gas fields



- These assets are considered natural resources while they are still part of the land
- As they are extracted from the land and converted into products, they are then accounted for as inventory (**raw materials**)



Depletion

B/S:

- Natural resources are recorded on the company's books like a fixed asset, at cost
 - with total costs including all expenses to acquire and prepare the resource for its intended use



I/S:

- As the resource is consumed (converted to a product), the cost of the asset must be expensed (income statement)
 - This process is called **depletion**

Fundamentals of Depletion of Natural Resources

- As with depreciation of non-natural resource assets, a contra account called **accumulated depletion**, which records the total depletion expense for a natural resource over its life, offsets the natural resource asset account.
- Depletion expense is typically calculated based on the number of units extracted from cutting, mining, or pumping the resource from the land, similar to the units-of-production method.

- For example, assume a company has an oil well with an estimated 10,000 gallons of crude oil.
- The company purchased this well for \$1,000,000, and the well is expected to have no salvage value once it is pumped dry.



- The depletion cost per gallon
= \$ / = \$
- If the company extracts 4,000 gallons of oil in a given year,
depletion expense = \$

LEARNING OBJECTIVES

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-  **1. Distinguish between tangible and intangible assets;**
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- 5. Describe some special issues in accounting for long-term assets.**

Fundamentals of Amortization of Intangible Assets

- Recall that intangible assets are recorded as long-term assets at their cost.
- As with tangible assets, many intangible assets have a finite (limited) life span so their costs must be allocated over their useful lives
 - This process is **amortization**
- Depreciation and amortization are similar in nature but have some important differences.



1. First, amortization is typically only done using the straight-line method.
2. Second, there is usually no salvage value for intangible assets because they are completely used up over their life span.
3. Finally, an accumulated amortization account is **not** required to record yearly expenses (as is needed with depreciation); instead, the intangible asset account is written down each period.

RESEARCH
RESEARCH
RESEARCH

- For example, a company called Patents-R-Us purchased a product patent for \$10,000, granting the company exclusive use of that product for the next twenty years.
- Therefore, unless the company does not think the product will be useful for all twenty years (at that point the company would use the shorter useful life of the product), the company will record amortization expense of:



- The amortization expense per year
= \$ _____ = \$ _____
- Assuming that it was placed into service on October 1, 2019, amortization expense for year 2019 will be:



JOURNAL			
Date	Account	Debit	Credit
Oct. 1, 2019	<i>To record amortization on patent for period</i>		

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Revision of Remaining Life or Salvage Value

Example:

- Kenzie has a press worth \$58,000
- Its salvage value was originally estimated to be \$10,000
- Its economic life was originally estimated to be five years
- Kenzie uses straight-line depreciation

After three years, Kenzie determines that the estimated useful life would have been more accurately estimated at eight years, and the salvage value at that time would be \$6,000. The revised depreciation expense is calculated as shown:

- Original cost
- Depreciation previously taken
- Book value at beginning of year 4
- Revised salvage value
- Revised remaining depreciable cost

- Revised remaining useful life
- Revised depreciation (straight-line method)

Sale of an Asset

- When an asset is sold, the company must account for its depreciation up to the date of sale.
- This means companies may be required to record a depreciation entry before the sale of the asset to ensure it is current.



1. Gain
2. Loss
3. Breakeven

Sale of an Asset: Calculating Gain or Loss

Example:

- Kenzie has a press worth \$58,000.
- Its salvage value was originally estimated to be \$10,000.
- Its economic life was originally estimated to be five years.
- Kenzie uses straight-line depreciation.
- Kenzie sells the press at the end of the third year:

Selling price = \$31,000

Cost of Press	
Less: Accumulated Depreciation: Printing Press	
<u>= Book Value</u>	

Sales Price
Less: Book Value
= _____ on Sale of Printing Press

Selling price = \$27,100

Cost of Press
Less: Accumulated Depreciation: Printing Press
= Book Value

Sales Price	
Less: Book Value	
=	on Sale of Printing Press

Sale of an Asset: Recording the Sale

Selling price = \$29,200
Gain = \$

JOURNAL			
Date	Account	Debit	Credit
Dec. 31, 2019			

Selling price = \$31,000
Gain = \$1,800

JOURNAL			
Date	Account	Debit	Credit
Dec. 31, 2019			

Selling price = \$27,100
Loss = \$2,100

JOURNAL			
Date	Account	Debit	Credit
Dec. 31, 2019			

Summary

Tangible versus Intangible Assets

Tangible assets:

- Tangible assets are assets that have physical substance.
- Long-term tangible assets are assets used in the normal course of operation of businesses that last for more than one year and are not intended to be resold.
 - Examples of long-term tangible assets are land, building, and machinery.

Intangible assets:

- Intangible assets lack physical substance but often have value and legal rights and protections, and therefore are still assets to the firm.
 - Examples of intangible assets are patents, trademarks, copyrights, and goodwill.

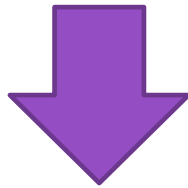
Summary

Capitalized Costs versus Expenses

Costs incurred to purchase a long-term asset

(Expenditures such as labor, services, materials are examples of items that are fully deducted from current year taxable income - “**EXPENSES**”)

Used in the day-to-day operations of the business?



Yes

Capitalized and then depreciated over the useful life of that asset

No

was purchased for investment purposes/will be considered an investment asset:

- short term (can be converted to cash in one year) or
- long term (held for over a year)

Summary

Apply Depreciation Methods to Allocate Capitalized Costs

Overview

- Fixed assets are recorded at the historical (initial) cost, including any costs to acquire the asset and get it ready for use.
- Depreciation is the process of allocating the cost of using a long-term asset over its anticipated economic (useful) life.
- To determine depreciation, one needs the fixed asset's historical cost, salvage value, and useful life (in years or units).

Methods

- There are three main methods to calculate depreciation:
 - 1) the straight-line method
 - 2) units-of-production method, and
 - 3) double-declining-balance method
- Natural resources are tangible assets occurring in nature that a company owns, which are consumed when used.
- Natural resources are depleted over the life of the asset, using a _____ method.

Summary

Accounting for Intangible Assets

With finite life:

- Finite intangible assets are typically amortized using the straight-line method over the useful life of the asset.
- This is similar to depreciation but is credited to the intangible asset rather than to a contra account.

With infinite life:

- Intangible assets with an indefinite life are not amortized.
- They are assessed yearly for **impairment**.

Summary

Some Special Issues in Accounting for Long-Term Assets

Adjustments to asset's useful life or to its SV:

- Because estimates are used to calculate depreciation of fixed assets, sometimes adjustments may need to be made to the asset's useful life or to its salvage value.
- To make these adjustments, the asset's net book value is updated, and then the adjustments are made for the remaining years.

Sale of an Asset:

- Assets are sometimes sold before the end of their useful life.
- These sales can result in a gain, a loss, or neither, depending on the cash received and the asset's net book value.

LEARNING OBJECTIVES

After this class, you should be able to

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Key Terms

1. accumulated depletion
2. accumulated depreciation
3. amortization
4. capitalization
5. contra account
6. copyright
7. current expense
8. depletion
9. depreciation
10. double-declining-balance depreciation method
11. fixed asset
12. functional obsolescence
13. goodwill
14. intangible asset
15. investment
16. long-term asset
17. natural resources
18. patent
19. physical obsolescence
20. salvage (residual) value
21. straight-line depreciation
22. tangible asset
23. trademark
24. units-of-production depreciation method
25. useful life
1. contra account that records the total depletion expense for a natural resource over its life
2. contra account that records the total depreciation expense for a fixed asset over its life
3. allocation of the costs of intangible assets over their useful economic lives; also, process of separating the principal and interest in loan payments over the life of a loan
4. process in which a long-term asset is recorded on the balance sheet and its allocated costs are expensed on the income statement over the asset's economic life
5. account paired with another account type, has an opposite normal balance to the paired account, and reduces the balance in the paired account at the end of a period
6. exclusive rights to reproduce and sell an artistic, literary, or musical asset
7. cost to the business that is charged in the current period
8. expense associated with consuming a natural resource
9. process of allocating the costs of a tangible asset over the asset's economic life
10. accelerated depreciation method that accounts for both time and usage, so it takes more expense in the first few years of the asset's life
11. tangible long-term asset
12. reduction of an asset's value to the company, not including physical obsolescence
13. value of certain favorable factors that a business possesses that allows it to generate a greater rate of return or profit; includes price paid for an acquired company above the fair value of its identifiable net assets
14. asset with financial value but no physical presence; examples include copyrights, patents, goodwill, and trademarks
15. short-term and long-term asset that is not used in the day-to-day operations of the business
16. asset used ongoing in the normal course of business for more than one year that is not intended to be resold
17. assets a company owns that are consumed when used; they are typically taken out of the earth
18. contract providing exclusive rights to produce and sell a unique product without competition for twenty years
19. reduction in the value of an asset to the company based on its physical deterioration
20. price that the asset will sell for or be worth as a trade-in when the useful life is over
21. depreciation method that evenly splits the depreciable amount across the useful life of the asset
22. asset that has physical substance
23. exclusive right to a name, term, or symbol a company uses to identify itself or its products
24. depreciation method that considers the actual usage of the asset to determine the depreciation expense
25. time period over which an asset cost is allocated



Questions?